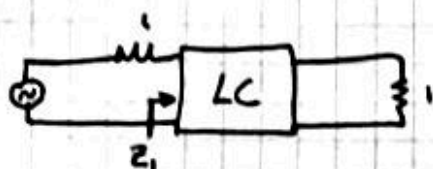


T5/3



$$S_{11} = \frac{Z_1 - R_0}{Z_1 + R_0} \quad \left\{ \quad Z_1 = R_0 \frac{(1 + S_{11})}{(1 - S_{11})} \right.$$

$$|S_{21}|^2 = |T(s)|^2 = \frac{1}{1 + \xi^2 C^2(n)}$$

$$\begin{aligned} C(n) &= 2\omega C_{n-1}(\omega) - C_{n-2}(\omega) \\ C_0 &= 1 \quad C_1 = \omega \\ C_2 &= 2\omega^2 - 1 \\ C_3 &= 4\omega^3 - 3\omega \end{aligned}$$

$$|S_{21}|^2 = \frac{1}{1 + \xi^2 (4\omega^3 - 3\omega)^2} = \frac{\frac{1}{\xi^2}}{\frac{1}{\xi^2} + 16\omega^6 - 24\omega^4 + 9\omega^2}$$

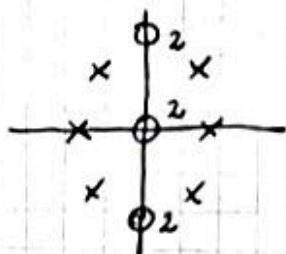
→ Método de Darlington

$$|S_{11}|^2 + |S_{21}|^2 = 1 \quad \rightarrow \quad |S_{11}|^2 = 1 - |S_{21}|^2 = \frac{16\omega^6 - 24\omega^4 + 9\omega^2}{\frac{1}{\xi^2} + 16\omega^6 - 24\omega^4 + 9\omega^2}$$

$$|S_{11}|^2 \Big|_{\frac{j}{\xi}} = S_{11}(s) S_{11}(-s) = \frac{-16s^6 - 24s^4 - 9s^2}{\frac{1}{\xi^2} - 16s^6 - 24s^4 - 9s^2}$$

$$\begin{aligned} \alpha &= 1 \text{ dB} \\ \text{con } \xi^2 &= 10^{\alpha/10} - 1 \\ \xi^2 &= 0.259 \end{aligned}$$

→ con Calculadora se tienen las raíces y se eligen polos y ceros del semiplano izquierdo



$$S_{11}(s) = \frac{s(s + j0.86)(s - j0.86)}{(s + (0.24 + j0.966j))(s + 0.4941)}$$

$$S_{11}(s) = \frac{s(s^2 + 0.77)}{s^3 + s^2 + 1.2307s + 0.4923}$$

$$\rightarrow Z_1 = \frac{1+s_{11}}{1-s_{11}} \rightarrow s_{11} = \frac{p}{Q} \rightarrow \frac{1+\frac{p}{Q}}{1-\frac{p}{Q}} = \frac{\frac{Q+p}{Q}}{\frac{Q-p}{Q}} = \frac{Q+p}{Q-p}$$

$$Z_1 = \frac{2s^3 + s^2 + 2s + 0.492}{s^2 + 0.461s + 0.492}$$

→ Síntesis de Z_1

$$\frac{2s^3 + s^2 + 2s + 0.49}{s^2 + 0.46s + 0.49} \quad \frac{2s}{2} \quad \underline{\hspace{1cm}}$$

$$\frac{s^2 + 0.46s + 0.49}{s^2 + 0.492s + 0} \quad \underline{0.078s^2 + 1.02s + 0.492} \quad \sim s \quad \frac{1}{1}$$

$$\frac{s + 0.49 \cdot 0.49}{s + 0} \quad \underline{s} \quad \frac{1}{0.49} \quad \underline{\hspace{1cm}}$$

$$\frac{0.49}{0.49} \quad \underline{0 + 0.49} \quad \frac{1}{0.49} \quad \underline{\hspace{1cm}}$$

$$\frac{0}{0} \quad \underline{1} \quad \frac{1}{1}$$

